

Supplementary material — v0.1

This supplement reproduces, verbatim and in English, the sealed method and result sections of the two internal manuscripts that the Korean main text merges: **Phase 1** = the three-archetype confirmatory experiment (internal Paper 1, draft v0.8, SHA-256 37f304b0a385...) and **Phase 2** = the condition-box campaign with its finite-batch identification correction (internal Paper 2, v0.4, SHA-256 d71f15dfb808...). Wherever the text below says "Paper 1" or "Paper 2", read Phase 1 or Phase 2 of the present study. Table and figure numbers inside each part are those of the source manuscript. Every table is regenerated from the sealed artefacts by the scripts named in the text; no number was retyped. Phase-1 tables carry the interpretation lock of the sealed protocol (synthetic archetype · nested composition path · hard-only · time-contract-conditional · stress-tail mixture; design/formal_interpretation_seal.md).

Contents: S1 Phase 1 methods · S2 Phase 1 results and discussion · S3 Phase 2 methods (design, stages, analysis plan, verification pack) · S4 Phase 2 results (labels, re-scoring, arithmetic comparator, band sensitivity, deviations, fitted surfaces, reader validation) · S5 Phase 2 discussion (identification limits, non-claims, multiplier scale).

S1. Phase 1 — methods (verbatim)

2. Methods

2.1 Design overview and pre-specification

The study is a computational experiment under a frozen protocol (nrf-v3.10). All scientific parameters — archetypes, headcount grids, replication counts, decision lines, censoring policy, and analysis plan — were fixed before the confirmatory campaign. Execution followed a staged pipeline: (i) a scripted software **qualification** battery; (ii) an operational **pilot** (297 evaluations) gating stability, error, timing and memory, whose cell rates are *not* used in scientific denominators; and (iii) the sealed **formal** campaign of 9,150 terminal evaluations reported here. A prior pilot under an earlier execution contract (v3.9) ended in an acceptance NO-GO due to time-censoring load; the execution contract (search mode and time ladder) was revised and fully re-verified as v3.10 rather than mixing results across contracts. Two earlier formal attempts were abandoned (one mid-run resource-contract change, one arm-implementation error caught by fail-closed error handling); both are retained as tainted history and the authoritative attempt (formal_attempt_003) started fresh.

2.2 Synthetic scenario generator

Three demand archetypes fix the daily D/E/N minimums: **A** 5/5/5 (headcounts $H = 23\text{--}26$), **B** 7/7/6 ($H = 30\text{--}34$), and **C** 9/9/8 ($H = 39\text{--}45$). The nominal 1.6 cells are A24, B32 and C42 (daily total $\times 1.6$, rounded up). For each archetype, staff composition follows a **nested composition path**: role mix

(manager/senior/junior/generic) and the number of monthly rest, annual-leave and education/duty beneficiaries are deterministic functions of H , so that moving from H to $H+1$ adds a specific role and can raise leave-event load. This models realistic scale-up of a full-time workforce, but it means a one-head increment is *not* a ceteris-paribus body count — a property central to interpreting adjacent contrasts (§3.4).

Each scenario root draws an independent synthetic month: leave/education events are placed under frozen rules, and the previous-month boundary is drawn from a sealed bank of feasible prior-month tails spanning four stress categories (neutral, work-boundary, evening-boundary, night-recovery) in a pre-specified stress mixture. The mixture is a design choice, not an empirical prevalence estimate.

Within an archetype, all headcounts share scenario roots, enabling paired one-nurse contrasts.

Archetype A's skeleton has historical provenance in anonymized operational logs, but every study value is researcher-set and synthetic; no operational data enter the analysis.

Relation to standard NRP benchmark sets. We deliberately do not use the public nurse-rostering benchmark sets — NSPLib (Vanhoucke and Maenhout, 2009) and the International Nurse Rostering Competition instances (Haspeslagh et al., 2014; Ceschia et al., 2019) — because they cannot carry this study's estimand. Those sets provide fixed instances, or generators parameterized by coverage/preference complexity indicators, for comparing rostering *algorithms* on a *given* workforce, typically under objectives dominated by soft-constraint costs. Our design inverts that layout: the

solver and rule contract are frozen, and the measured quantity is the probability that *any* valid month exists as the integer headcount sweeps a pre-specified grid over a *distribution* of synthetic months. This estimand requires (i) thousands of independent month roots drawn from a frozen generator, not a fixed instance list; (ii) a nested composition path linking role mix and leave entitlements to headcount; (iii) a hard-only monthly contract — monthly night caps, minimum off-days, night-recovery, forbidden transitions, role/novice coverage, and prior-month continuity via sealed feasible tails — that the benchmark formats do not encode; and (iv) common-random-number coupling of roots across headcounts and arms for paired contrasts. Conversely, this study makes no algorithmic performance claim, and its results are not comparable to benchmark optimization scores. Positioning the three archetypes within NSPLib-style complexity-indicator space, or constructing encoding-compatible bridge instances, is left to the successor study.

2.3 Hard-constraint model and solver

Feasibility is evaluated with a hard-only constraint satisfaction model on Google OR-Tools CP-SAT (version **9.12.4544**). The model enforces the full monthly hard-rule contract — daily role/shift coverage, forbidden transitions, consecutive-work and consecutive-night limits, night-recovery rules, monthly night caps (≤ 8), an **exact** monthly count of pure off-days (9 per non-manager; approved leave and education days are additional non-available days, not substitutes), role/seniority coverage, and prior-month continuity — and **excludes** soft preferences, fairness objectives, overtime,

float/agency staff, and post-publication swaps. The information basis is pre-publication: only what is known when a roster is posted (staffing, approved leave, education, prior-month end state) enters the model; post-publication disruptions and recovery actions are a different estimand reserved for successor studies. The solver runs as a satisfaction (not optimization) model with a fixed random seed and two workers per solve; the first valid schedule (if any) is checked by an independent validator before an F verdict is recorded.

2.4 Time contract and terminal states

Each evaluation ascends a deterministic-time ladder (1, 10, 100, 200 solver deterministic-time units, with wall-clock watchdogs of 120/600/1800/1200 s) until the solver returns a definite answer or the ladder ends. Terminal states are:

State	Meaning	Analysis use
F	Valid monthly schedule found and validated	feasibility indicator
I	Infeasibility proved under the model	model-conditional non-feasibility
T	Time-censored at ladder end	never imputed to F or I; widens identification
E	Execution/contract error	attempt fail-closed; no partial replacement

T is reported as censoring: cell-level probabilities are partially identified in $[F/n, (F+T)/n]$. Any E terminally taints an attempt (no selective re-runs); the authoritative attempt completed with $E = 0$.

2.5 Experimental plan

The formal campaign comprised 9,150 evaluations: the three primary 1.6 cells at $n = 1,350$ roots each (4,050), thirteen secondary headcount cells at $n = 300$ (3,900), a paired +2 maximum-staffing cap arm on 300 roots at each primary headcount (900), and a paired night-demand arm (B32, 7/7/6 $\rightarrow$ 6/7/7) on 300 roots (300). Evaluations were staged (main $\rightarrow$ cap2 $\rightarrow$ night) across a fixed three-node pool with frozen per-node concurrency ceilings. 9,150 is an evaluation count, not an independent-ward count; independent roots number 4,050.

2.6 Statistical analysis (pre-specified)

Primary family. For each 1.6 cell ($n = 1,350$), simultaneous Clopper–Pearson bounds at $\alpha_{\star} = 0.05/6$ give a lower bound L on F/n and an upper bound U on $(F+T)/n$. The cell **meets** the 95% research benchmark if $L \geq 0.95$ (equivalently $F \geq 1,302$), **does not meet** if $U < 0.95$ ($F+T \leq 1,262$), and is **indeterminate** otherwise. The benchmark is a research reference, not a clinical sufficiency criterion.

Censoring sensitivity. T/n is reported for every primary cell; a pre-specified rule withdraws sample-size/power claims if any primary cell shows $T/n > 0.005$ (decisions themselves always follow the $F / F+T$ rule).

Secondary analyses. The 16-cell curves and 13 adjacent paired contrasts (risk differences on shared roots, with conservative outer confidence intervals under censoring) are descriptive and not multiplicity-controlled. A pre-specified audit checklist governs any non-monotone narrative (§3.4). The

cap2 gate pre-commits title/abstract limitations if the cap arm shows a large feasibility loss

(conservative RD upper bound < -0.05 or cap-arm F rate < $\frac{1}{2}$).

2.7 Reproducibility, sealing and audit

The generator, model contract, seeds, and analysis thresholds are frozen and hash-pinned; per-evaluation records are sealed at completion, and a built-in audit subcommand re-verifies the seal hash, the execution-contract hash, and per-stage terminal counts and gates. The primary decisions are recomputed from the sealed records by the packaged analysis scripts, which share the study's own decision code: this is self-implementation reproduction, **not** third-party audit. Analysis tables and figures in this paper are regenerated by scripts included in the study package. Aggregated tables, figures and analysis code are shared; per-evaluation sealed records and the pinned commercial rostering product source are retained institutionally (see Data and code availability).

2.8 Ethics

Human-subjects research: not applicable — the study uses fully synthetic scenarios with no patient, staff, or identifiable data.

S2. Phase 1 — results and discussion (verbatim)

3. Results

3.1 Execution

The sealed formal attempt completed 9,150/9,150 evaluations with $E = 0$ in ≈ 1.8 h of wall time on the three-node pool. Overall terminal states were $F\ 8,537 \cdot I\ 584 \cdot T\ 29$. All three stages passed their gates (main 7,950: $F\ 7,412/I\ 510/T\ 28$; cap2 900: $F\ 844/I\ 56/T\ 0$; night 300: $F\ 281/I\ 18/T\ 1$). The seal-integrity audit passed (seal hash, contract hash, per-stage counts and gates).

3.2 Primary confirmatory decisions (Table 1, Figure 1)

Cell	H	F	I	T	F/n	L	U	Decision
A 1.6	24	1,287	63	0	0.9533	0.9378	0.9660	indeterminate
B 1.6	32	1,262	88	0	0.9348	0.9170	0.9499	does not meet
C 1.6	42	1,226	124	0	0.9081	0.8877	0.9260	does not meet

No primary cell met the pre-specified 95% benchmark under simultaneous exact inference. A24's point estimate lies just above 0.95 but its simultaneous lower bound does not; B32 and C42 fall below by upper bound. Descriptively, the feasibility rate at the nominal multiplier declines across archetypes as demand scale grows ($0.953 \rightarrow 0.935 \rightarrow 0.908$); because archetypes are design benchmarks rather than a sampled population, we do not attach a causal scale interpretation. All primary cells had $T = 0$, so the pre-specified power-withdrawal rule was not triggered. We flag one margin explicitly: B32's $F + T = 1,262$ sits exactly on the pre-specified does-not-meet line (at $n = 1,350$ the rule fires at $F + T \leq$

1,262), with $U = 0.9499$ — a single additional feasible month would have returned an indeterminate call. C42 ($F = 1,226$) lies 36 months inside the line, and A24's indeterminate call is 15 months from the meets line ($F \geq 1,302$). The three primary calls are therefore not equally robust to sampling variation.

3.3 Descriptive feasibility curves (Table 2, Figure 2)

All 16 cells completed ($E = 0$). F rates ranged from 0.857 (B30, with $T = 26/300$ — the only high-censoring cell; identification band $[0.857, 0.943]$) to 0.977 (A25). Within the tested grids, no archetype's curve reaches a plateau at 1.0; C remains between 0.908 and 0.950 across $H = 39$ –45 (multiplier ≈ 1.50 –1.73). Secondary cells ($n = 300$) carry $\approx \pm 2.5$ pp Monte Carlo uncertainty and are not held to the primary decision lines.

3.4 Adjacent one-nurse contrasts and non-monotonicity (Table 3, Figure 3)

Of 13 adjacent paired contrasts, the F rate was non-increasing in 5 (largest: C41→42, identified RD -0.043 ; also A25→26, B31→32, B33→34, C43→44). The pre-specified audit found the flips to be almost entirely $F \rightarrow I$ transitions on shared roots (e.g., C41→42: 16 $F \rightarrow I$ vs 3 $I \rightarrow F$), not censoring artifacts; error states were absent; and tail stress class was stable within roots across H . The direction is consistent with the nested composition path: adding one head changes the role prefix and raises leave-event load under the frozen scaling rules. We therefore report non-monotonicity **descriptively** as a property of the nested path, and do not claim that an additional nurse per se reduces feasibility, nor that one always helps.

3.5 Structural sensitivities (Tables 4–5, Figure 4)

Maximum-staffing cap (+2). On 300 shared roots per archetype at the primary headcounts, cap2 terminal outcomes matched the main arm exactly (identified paired RD = 0 in all three archetypes).

The pre-committed cap2 limitation gate was not triggered; primary results are robust, on this sample, to capping daily over-assignment at minimum + 2.

Night-demand shift. Moving one daily slot from day to night (B32, 7/7/6 → 6/7/7; n = 300 pairs) produced a negligible paired change (identified RD $\approx$ [−0.003, 0]; outer CI $\approx$ [−0.021, 0.015]). This is a local structural sensitivity; 6/7/7 is not a clinical standard demand pattern.

4. Discussion

4.1 Principal findings

Under a sealed synthetic hard-only contract, the nominal 1.6 multiplier's headcounts did not earn a "meets the 95% benchmark" call in any of three pre-specified demand archetypes — one cell was indeterminate and two fell short — and the multiplier's position on the feasibility curve varied by archetype. One-nurse increments did not monotonically improve feasibility under the nested composition path. Feasibility was insensitive, on paired samples, to a +2 staffing cap and to a one-slot night shift.

4.2 What the multiplier result does and does not mean

The allowed reading is conditional: *within the pre-specified synthetic archetypes and composition path, the 1.6 arithmetic total sits at the measured positions on each conditional feasibility curve*. This is a method-validation result about slot-arithmetic heuristics as monthly-feasibility predictors under one frozen contract — not evidence that any real ward is understaffed, that 1.6 is universally wrong (or right), or that some other multiplier is optimal. What the differentiated primary decisions do establish is that a single fixed multiplier cannot be assumed to deliver uniform monthly-roster feasibility across demand conditions even in a clean synthetic setting; condition-resolved estimation is required.

4.3 Non-monotonicity and the nested path

The audited $F \rightarrow I$ flips at one-head increments illustrate a structural point for staffing arithmetic: when workforce scale-up follows realistic composition rules (roles and leave entitlements growing with headcount), the *effective* schedulable capacity need not grow with nominal headcount at every step. Slot-arithmetic necessary conditions are satisfied with margin at all primary cells, yet feasibility still falls short of the benchmark — residual infeasibility is combinational, arising from the interaction of role coverage, leave placement, transition rules and continuity, not from monthly slot shortage.

4.4 Post-hoc exploratory diagnostics (off the primary contract)

Two post-hoc diagnostics, run after sealing on read-only records and labeled exploratory, sharpen the interpretation without altering any sealed table. First, extending the time ladder (400/800 units) on the 26 censored B30 roots resolved 16 to feasible and 3 to proven-infeasible, leaving 7 censored — the

censored pool is a mixture of latent feasibles, latent infeasibles and a hard core, supporting the partial-identification treatment of T rather than any imputation. Second, a decoupling probe on the 16 C41→42 F→I roots re-ran H = 42 with role mix and leave load frozen at the H = 41 path plus one generic staff member: the nested path reproduced infeasibility on 16/16 roots while the decoupled arm recovered feasibility on 16/16, experimentally confirming that the paired drop is a composition-path effect rather than a headcount effect. Both diagnostics are hypothesis-generating for the successor study and make no confirmatory claims.

4.5 Strengths

The design commits to decisions before data (pre-specified cells, replication counts, decision lines, and a censoring-sensitivity rule); treats solver censoring as first-class data; separates software qualification and operational piloting from scientific denominators; enforces fail-closed error handling with no selective replacement; and seals records so that the decisions can be recomputed from them (by the study's own analysis scripts; no third-party audit was performed). Reporting follows simulation-study reporting guidance where applicable [TBD: STRESS-adjacent checklist citation].

4.6 Limitations

Results are conditional on (i) three synthetic archetypes that represent design benchmarks, not sampled wards; (ii) a hard-only model that excludes soft preferences, fairness, overtime, float and swaps; (iii) a frozen search/time contract — I and T are defined relative to it; (iv) a pre-specified stress-

tail mixture for prior-month continuity whose real-world prevalence is unknown; (v) a nested composition path that deliberately couples headcount with role/leave structure; and (vi) no linkage to the public NRP benchmark sets (NSPLib, INRC), whose instance formats cannot express this monthly hard-rule contract (§2.2) — cross-benchmark positioning remains future work; (vii) a pre-specified annual-leave and education load (≈ 0.2 leave days per nurse-month by design, ≈ 0.3 including education) that is **below** typical domestic leave utilisation — a heavier leave load raises the arithmetic necessary condition itself, because the same demand must be covered from fewer available nurse-days, so the reported feasibility rates are, if anything, optimistic relative to wards with higher leave load; and (viii) the protocol was **internally pre-specified with no external timestamp** — no public registration record exists, and none was created after the results were known. Feasibility here does not imply fairness or acceptability of any produced schedule, and the first-found schedules' load distributions should not be read as fairness outcomes. External validity to real wards is a matter for future studies with real data; none is claimed.

4.7 Non-claims

This study does not establish safe, minimum, adequate or optimal staffing for any real unit; does not validate or refute the 1.6 rule in general; does not equate model infeasibility with real-world impossibility; and does not treat 9,150 evaluations as 9,150 independent wards (independent roots: 4,050).

4.8 Future work

The measurement contract validated here — generator, four-state taxonomy, partial identification, sealed records with an internal integrity check — is the anchor for a successor study estimating the feasibility boundary surface $H^*(x; \tau)$ over a parameterized condition space, with decomposition into an arithmetic necessary condition $N_0(x)$ and a combinatorial premium $\Delta = H^* - N_0$, and an accuracy–simplicity frontier of portable staffing formulas. The τ -sensitivity of the present decisions and the demonstrated path-dependence of one-head effects (Section 4.4) directly motivate treating the reliability benchmark τ and the composition path as explicit axes of that surface, and the leave-load limitation (vii) motivates a third: annual-leave and discretionary-leave load, which shifts the arithmetic necessary condition rather than only the combinatorial premium.

5. Conclusion

Under a sealed, pre-specified synthetic contract, fixed-multiplier staffing arithmetic did not guarantee monthly-roster hard-feasibility at a 95% research benchmark in any of three demand archetypes, and its adequacy varied across conditions. Monthly feasibility is a combinatorial property that daily slot arithmetic — and any single multiplier — cannot fully capture. The study's sealed, censoring-honest measurement contract provides a reproducible foundation for mapping condition-resolved staffing feasibility boundaries.

S3. Phase 2 — methods (verbatim)

2. Methods

2.1 Design overview and honest pre-specification status

The study is a computational experiment under a frozen execution contract (nrf2-r2-execution-contract, design pin *Research 2.0* v1.0.17, contract v1.3 self-hash 9efe0e9ca009...). The contract fixes the original estimand language, τ , the stair, per-cell sample sizes, the stage sequence, the budget caps and nine automatic stopping conditions (S1–S9) — one terminal E halts the campaign; a budget cap halts before the request that would exceed it; time-censoring is never a halt or a gate; a completed stage's gate is mechanical (record counts, key uniqueness, contract hash, validator agreement) and contains no result-dependent criterion. Execution was approved once, before any solve under the contract; stages report and proceed, and no re-approval is sought mid-campaign. Every later amendment is recorded in the contract's approval.amendments with its reason and its effect on the stopping-condition hash (3ae8c579... at v1.3).

Pre-specification is **internal, with no external timestamp**. Repository records document the contract and promotion-rule freezes; the Stage C selection rule was fixed before aggregation of the existing map records, and the Stage D selection rule before Stage D execution. These are distinct timing claims: the Stage C freeze did not precede production of all map data. No external registry was used. These internal timestamps do not establish external registration or complete blinding.

The GAM prediction table and Stage D run freeze were committed before Stage D began. The M1/M2 formula ladder and its reader were committed while Stage D was running, using map-plus-Stage C training inputs. The operational monitor exposed F/I/T summaries for x347 h4, x348 h0 and x348 h1; project records state that these three monitor cells were used for throughput estimation, not fitting. Nothing in that disclosure converts the holdout into a registered, fully blinded trial.

The stage sequence is: a self-contained verification pack (§2.12) → Stage 0a (compiler/adaptor verification on 560 job pairs) → Stage 0b (a seven-condition pilot with a pre-specified two-rung extension) → Stage B screen (300 Sobol points, two waves) → Stage B map (promoted windows at $n = 500$) → Stage C infill (contract v1.2) → analysis freeze of the prediction table → Stage D holdout (contract v1.3). Every stage ran to its mechanical gate. Reporting follows the STRESS guidelines for empirical simulation studies (Monks et al., 2019) where their items apply. The completion-set reader of §2.6 is a post hoc descriptive correction of interpretation. It does not refit models, rerun solvers, change τ , or assert a population confidence bound.

2.2 Condition space

A ward condition is an eight-scalar vector x . The box is read from code (AXIS_BOX), not typed. Table 1 records the axes, ranges and Sobol mapping.

Table 1. Condition space and screen design.

axis	meaning	range
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W	weekday daily demand total $D + E + N$ (people on duty per day)	12–27 (integer)
nu	night fraction $v = N / W$	0.25–0.4
rho	evening-to-day ratio $\rho = E / D$	0.85–1.15
OF	exact monthly count of pure off-days per non-manager	8–12 (integer)
P	holiday-type designated days in the month	0–5 (integer)
lambda_l	annual-leave load λ_L , employee-days per productive person per month	0.5–1.2
j	junior (novice) fraction of productive staff	0.05–0.3
s	senior fraction of productive staff	0.25–0.5

Read from `src/nurse_feasibility_nrf2/stage_b.py` (AXIS_BOX, AXIS_ORDER). Sobol coordinates map affinely onto the float ranges and by $\text{floor}(u \cdot n)$ onto the integer ranges; the axis order is the design's contract with the unit cube. Stair rungs $h = 0\text{--}8$ above the arithmetic floor $N_0(x)$. Prior-month tail bank per category: screen 1, map 4.

Fixed across the space: a 30-day month, one manager, full-time staff, and the twenty hard rules, solver parameters and time ladder of Paper 1's protocol `nrf-v3.10`, inherited read-only. The annual-leave load λ_L is the primary axis of leave stress; the two sentinel values used in Paper 1 (0.2, anchor position) and in the pilots (2.0, stress) are outside the Sobol flood. The optional night-cap sensitivity arm of the design ($k_N = 7/8/9$) was not run. W is externally specified daily staffing demand (people on duty per weekday), not a demand-estimation model. The output is roster headcount under the declared

synthetic rules. It is not automatically annual FTE, a legal staffing standard, or an evaluated patient-safety threshold.

Sobol points are a fixed low-discrepancy design on the unit cube, mapped affinely onto the box. They are not a random sample of real wards, and neither the 540 promoted stairs nor the Stage D selected singleton subset is such a sample.

2.3 Scenario generator, composition arms and the arithmetic floor N_0

Each condition x and headcount H compiles to a monthly case: daily D/E/N minimums from W , v and p with a deterministic rounding rule; a roster of H people with one manager and a productive body whose junior and senior fractions are j and s ; rest, leave and holiday-type beneficiaries counted per the Paper-1 scaling with leave in employee-days set by λ_L ; P holiday-type days placed by a keyed rule; and a previous-month tail (§2.8). Two composition arms are run for every point:

- **nested** — the role mix and the beneficiary counts are functions of H , so that $H \rightarrow H+1$ adds a specific role and can raise leave load (Paper 1's path; one head is not a ceteris-paribus body);
- **decouple** — the role mix is frozen at the reference headcount and every additional head is a generic productive nurse, isolating the pure headcount effect. The decouple arm is the primary arm for the original H^* reading; the nested arm is the anchor and sensitivity arm.

$N_0(x)$ is computed by a solver-free preflight from three counting conditions — gross work-slots

(productive people $\times$ maximum work-days minus leave and holiday employee-days must cover $30 \times$

W), night-cap slots (productive people $\times$ min(available days, 8) must cover $30 \times N$) and role coverage — and the roster is probed for the ceiling at which the frozen role map can still be built. The compiler uses N_0 as a solver-free counting floor. Absence of recorded F below N_0 is an execution consistency check, not an additional proof over all unobserved headcounts. The response axis everywhere is $\mathbf{h} = \mathbf{H} - \mathbf{N}_0(\mathbf{x})$. Domain $h \geq 0$ is an inherited working N_0 premise. This paper does not prove that every unprobed roster above or beside the compiled floor is admissible, and it does not use that unproved claim to drop candidate rungs.

2.4 Hard-rule contract, solver, time contract and terminal states

Feasibility is decided by the same hard-only CP-SAT satisfaction model as Paper 1 (Google OR-Tools CP-SAT version 9.12.4544, pinned in the contract; two workers, fixed seed), invoked read-only from the pinned product commit through an adapter whose case format (M1–M6) is derived from the model builder's own parser. The rule contract — daily role/shift coverage, forbidden transitions, consecutive-work and consecutive-night limits, night recovery, the monthly night cap, an **exact** monthly count of pure off-days per non-manager (OF), role and seniority coverage, and prior-month continuity — is published as a machine-readable specification independent of the product code (output/spec_package/, RULE_CONTRACT v1.0.2), and an extractor fails when the specification and the sealed code disagree. Every F is confirmed by an independent validator that shares no constants with the solver; the OF equality is additionally recomputed by the driver from every stored schedule.

Each solve ascends the inherited deterministic-time ladder and ends in exactly one terminal state: **F** (valid month found and validated), **I** (infeasibility proved), **T** (time-censored at the ladder's end), **E** (execution error). T is censoring, not a requirement of any gate (lesson L1 of the programme: the anchor replication's overall NO-GO came from a $T = 0$ requirement that this design removed from the campaign gates). E is fail-closed: one terminal E halts the campaign. Integer count comparisons in the completion reader use exact rational arithmetic and $\geq$ at equality. Stored rounded whole-person M0/M1/M2 predictions are used unchanged; original rounding follows Python round. The $\text{ceil}(1.6 \text{ W})$ comparison is explicitly post hoc.

2.5 Response axis, two probabilities and two band constructions

Per cell (x, arm, h) the original analysis reported the decided proportion $\hat{p} = F/(F+I)$ with a Wilson interval (Wilson, 1927). On the screen, the crossing rung h^* was the smallest h whose $\hat{p}$ reached $\tau = 0.95$; on the map, a window $\{h^*-1, h^*, h^*+1\}$ was re-measured at $n = 500$ and classified by a local adjacent-upward-crossing rule. After Stage C, every rung of a stair was pooled over its map draw and its Stage C draws and re-read by the same local rule. That local rule does not implement a global minimum when cell rates are nonmonotone or lower rungs are missing. Treating its output as $H^*(x; \tau)$ was the principal interpretive error of v0.3.

Two probability targets remain distinct. Conditional q is estimated only from decided F and I. The all-generated target concerns feasibility of every generated case, including those whose solver outcome

was T; a denominator change alone does not identify their unobserved feasibility outcomes. Equality of the two is an assumption, not a finding. Two band constructions are therefore reported throughout §3:

- **Conditional mode.** A decided cell has band $[F/(F+I), F/(F+I)]$; a cell with no decisions has $[0, 1]$.
- **Censoring mode.** A nonempty cell has band $[F/n, (F+T)/n]$, $n = F+I+T$; a cell with no cases has $[0, 1]$. Synthetic T due to missing prior-month tails remains in n when present in the sealed pooled table.

These bands express logical missing outcomes in a finite batch. They are not 95% confidence intervals, and they are not simultaneous surface bands.

2.6 Completion-set reading of finite-batch cells

A rung is *certainly below* τ if its upper endpoint is below τ and *certainly attaining* if its lower endpoint is at least τ . A possible first attainment is any rung not certainly below τ that has no earlier certainly attaining rung. Missing rungs remain possible: holes in the candidate set are preserved, because an interval envelope alone can mis-score within-one-person hits. If there is no certainly attaining rung, the reader retains an unbounded unresolved tail and possible nonattainment. No finite top-of-ladder surrogate is inserted. Direction below all finite candidates is reported only conditional on a finite minimum existing; when no finite candidate remains, "below" refers to the whole unbounded tail. No numerical error is assigned to nonattainment.

Premises. (i) The headcount offset h is an integer on the working domain $h \geq 0$ (§2.3). (ii) Every rung h carries a closed band $[L_h, U_h] \subseteq [0, 1]$ from §2.5; a rung that was never observed carries the vacuous band $[0, 1]$, so it is neither certainly below nor certainly attaining. (iii) A *completion* assigns one number $p_h \in [L_h, U_h]$ to every rung; its first attainment is $\min\{h : p_h \geq \tau\}$, or none. (iv) $\tau = 19/20$ is compared in exact rational arithmetic and equality counts as attainment, so a rung with $F/(F+1) = 285/300$ attains. (v) No relation between rungs is imposed: shared tails, adaptive sampling and monotonicity never shrink the set.

Set. Let $B = \{h : U_h < \tau\}$ (certainly below) and $A = \{h : L_h \geq \tau\}$ (certainly attaining). The completion set is

$$C = \{h \geq 0 : h \notin B, \text{ and no } h' < h \text{ lies in } A\}.$$

If A is non-empty, C is finite and its largest element is $\min A$; if A is empty, C contains every rung outside B , continues beyond the last observed rung as an unbounded tail, and nonattainment remains possible. C is exactly the set of first attainments over all completions: if a completion first attains at h , then $p_h \geq \tau$ forces $U_h \geq \tau$ and each earlier $p_{h'} < \tau$ forces $L_{h'} < \tau$, so $h \in C$; conversely, for $h \in C$, the completion with $p_h = U_h$, $p_{h'} = L_{h'}$ for all $h' < h$ and arbitrary later values first attains at h . A singleton $C = \{\min A\}$ is the only case in which the finite-batch bands pin one first-attaining rung. Nothing in this construction asserts that the generator probabilities lie inside the bands; that is a sampling question the reader does not answer.

Procedure. (1) Form the band of each observed rung under the chosen mode; (2) mark A and B; (3) scan $h = 0, 1, 2, \dots$ up to $\min A$, or to the last observed rung when A is empty, keeping every rung not in B; the tail beyond the last observed rung is kept when A is empty. The implementation was unit-tested against all $3^4 = 81$ status patterns (missing, certainly below, certainly attaining) of a four-rung stair, each compared with brute-force enumeration of the 2^4 attain-or-not completions of that stair.

Worked examples (Stage D, conditional mode, $h = H - N_0$). Condition x318 ($N_0 = 31$) observed $h = 1, 2, 3$ with $F/(F+I) = 292/300, 282/300, 292/300$: $h = 1$ and $h = 3$ are certainly attaining and $h = 2$ is certainly below. The local adjacent-upward-crossing rule reported $h = 3$ ($H = 34$) because it looked for a rise from below τ ; the completion set is $\{0, 1\}$, because $h = 0$ was never observed and $h = 1$ already attains, so $h = 3$ cannot be a first attainment. Condition x350 ($N_0 = 34$) observed $h = 0, 1$ at $287/300, 280/300$: the v0.3 vocabulary called the stair "unresolved" because the rate fell after $h = 0$, whereas the completion set is the singleton $\{0\}$. Condition x300 ($N_0 = 36$) observed only $h = 9, 10, 11$, all certainly below τ ($279/300, 280/300, 279/300$); the completion set keeps the unobserved lower rungs $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$ and an unbounded tail from $h = 12$, so neither a finite minimum nor nonattainment is excluded.

The working domain remains $h \geq 0$. Cross-rung dependence induced by shared tails is not used to shrink the set. The resulting object is a conservative logical possibility set under the working domain,

not a sharp population identified set. Re-scoring frozen predictions against that set does not repair invalid training labels (Sections 2.11 and 3.3).

Controlled validation under designed truth. Because the sets are logical, not statistical, their

behaviour under sampling error was measured separately, without the solver

(scripts/nrf2_reader_validation.py; results in §3.9). The truth is a designed per-rung probability curve on

$h = 0\text{--}12$ — a step ($0.90 \rightarrow 0.97$), a plateau within one standard error of τ ($0.945 \rightarrow 0.952$), a dip after

attainment ($0.97, 0.93, 0.97$), a ramp rising 0.006 per rung through τ (clipped to $[0.90, 0.99]$), and a

curve that never attains (0.94) — with the true first attainment at $h^* \in \{0, 3, 6\}$. Each replication draws

a Stage D-like batch: a three-rung window centred at $h^* + \text{offset}$ for offsets $-2 \dots +2$ (a centre below

the floor is raised to $h = 0$, giving the two-rung floor window of §2.10), $n = 300$ cases per rung, each

case censored with probability 0, 0.02 or 0.10 independently of feasibility and otherwise decided F

with probability p_h , so that conditional and all-generated targets coincide by design. The

completion-set reader (conditional, censoring, and the censoring band widened by the §3.6 epsilons)

and the v0.3 local adjacent-upward-crossing rule are imported unchanged from the sealed erratum

code and from the map reader. Coverage means that the true first attainment lies in the completion

set, or, for the never-attaining curve, that nonattainment is kept possible; a singleton is *correct* when

its one rung is the true one; the local rule is *located and correct* when it names the true rung. 2000

replications per configuration, fixed seed. Designed curves are not ward populations, and no result of this check is a coverage statement about the campaign's own cells.

2.7 Stage 0: pilots and the nonmonotone property audit

Stage 0a executed 560 job pairs (F 970 / I 86 / T 64, E = 0) to verify the compiler, the adapter, the tail factory and the validator end-to-end under the contract. Stage 0b ran seven conditions $\times h = 0-6 \times$ both arms $\times n = 50$ (2,450 pairs; F 4,349 / I 348 / T 203) with a selection rule committed before Stage 0a's records were read, and its pre-specified two-rung extension added $h = 7, 8$ on the six conditions whose roster ceiling allowed it (600 pairs; F 1,118 / I 82 / T 0). The pilots' pre-specified property audit — descriptive, creating no threshold — found that the feasible rate does **not** climb monotonically in h : 33 of 84 one-rung steps decreased (39 %) in Stage 0b and 43 of 108 (40 %) with the extension, with 9 of 14 stairs assigned the historical flat-within-noise label, and the two leave-load levels 0.2 and 0.8 had observed rates ($\hat{p}$ 0.923 vs 0.929, overlapping Wilson intervals); these historical summaries do not establish equivalence or flat population curves. A further 550-pair probe on eleven unused conditions at $h = 8$ (F 1,026 / I 74 / T 0) found a residual shortfall on those additional conditions; this observation does not prove condition-invariance of the population probability. Two consequences were adopted as the *next* stage's pre-specification, not as a criterion change: the GAM's monotonicity constraint on $f(h)$ was removed (monotonicity is reported as a result, not imposed), and $\tau = 0.95$ was kept.

Nonmonotone finite-batch rates are therefore not an unexpected defect discovered after the holdout.

They are an empirical feature of these batches. Regardless of whether a population curve is monotone, the observed nonmonotone rates cannot justify relabelling a local crossing as the first global attainment.

2.8 Stage B: Sobol screen, promotion rule and map

Screen. 300 points of a scrambled Sobol sequence (Joe–Kuo direction numbers — Joe and Kuo, 2008 — with a derived digital shift; namespace `r2-v1.0|stage-B-screen|sobol-v1`) were mapped onto the box. Each point was run on both arms over the nine-rung stair $h = 0\text{--}8$ at $n = 120$ per cell, in two waves cut as prefixes of the Sobol sequence (179 and 121 points) so that a wave is never a cut across h .

Promotion rule (frozen before Stage B produced any observation). Per (point, arm): the smallest h whose decided screen proportion reaches $\tau = 0.95$ is the crossing; that rung and its two neighbours, clipped to the stair, are promoted to the map at $n = 500$ on that arm only. A stair with no crossing is recorded `boundary_not_located` and passed to Stage C's secondary level; a stair whose every cell is time-censored is recorded `fully_censored` and not promoted. The no-crossing branch was specified from Stage 0b's measurement (4 of 14 stairs never crossed) rather than left to judgement.

Map. The promoted windows were solved at $n = 500$ with a per-cell tail bank of 16 (§2.9). The map's completion gate is mechanical (pair count, duplicate keys, manifest keys, one arm per pair, τ unchanged, contract hash, $E = 0$, condition round-trip, no F below N_0 , validator agreement).

Promotion counts and screen-crossing tallies are reported in Table 3. Promotion selected stairs that crossed on the screen; it did not produce a probability sample of the box, and it did not fill missing lower rungs needed for a global minimum.

2.9 Prior-month tails: bank policy and dependence

Every case needs a feasible previous month whose end state constrains the new month's start. Stages 0a and 0b generated one tail per case per arm. Stage B instead uses a **bank**: per cell and tail category (neutral, work-boundary, evening-boundary, night-recovery), one tail on the screen and four on the map, sized so that the number of instances sharing one tail matches Paper 1's realised reuse (128 sealed tails over 4,050 instances, ≈ 31.6 per tail): 30.0 at screen grade and 31.25 at map grade. Stage C inherits the map policy.

The cost of banking is stated rather than smoothed over. A root reaches the tail through only two channels (the previous month's holiday selection and the sentinel), but those channels are nearly saturated: in one screen cell swept exhaustively, 120 roots produced 93 distinct holiday sets, 32 distinct sentinels and 117 distinct (holiday, sentinel) pairs. Banking therefore collapses real prior-month variation to a chosen resolution, and **instances sharing a tail also share a prior-month input**.

Conditional independence given a bank and marginal independence over bank generation are different assumptions. Per-cell Wilson intervals do not propagate tail-bank uncertainty; their nominal coverage for the intended generator population has not been established. No sign or magnitude of a

coverage error is estimated here. Resampling already-fixed stair labels does not propagate cell uncertainty through boundary reading. Stages 0a/0b (per-case tails) and Stages B/C/D (banked tails) differ in this respect, and pilot and map proportions are not pooled.

2.10 Stage C infill and Stage D holdout, with corrected freeze timing

Stage C and D were specified on 2026-09-02, while the map was running and before any map record had been read, and their selection rule was committed as a pure function of the map records, the screen summary and the contract before the records were aggregated. Applied mechanically to each promoted window at $\tau = 0.95$ (h_c below denotes the historical local crossing, not the population H^*):

- **R (located)** — a rung at or above τ whose next-lower rung is decided and below τ : eligible for precision infill (C-4: h_c and h_c-1 at $n = 500$ each, ordered by the crossing rung's margin $|p^* - \tau|$, until the budget is spent);
- **U (above or absent)** — every decided rung below τ : one upward extension, h_c+2 and h_c+3 , at $n = 500$;
- **L (at or below)** — every decided rung at or above τ : one downward extension, h_c-2 and h_c-3 ($h \geq 0$), at $n = 500$;
- **M (unresolved)** — a mixed window with no upward crossing, the non-monotone case the pilots measured; the rule treats it as R (no extension; eligible for precision infill at the smallest decided

rung reaching τ) and reports it as its own class, because the three-type scheme of the original specification is not exhaustive when $\hat{p}$ is not monotone.

Extension rungs that the roster ceiling cannot build are excluded by a probe, not by a name list, and re-derived at every freeze. Screen stairs that never reached $\tau = 0.95$ are **not** given a wider window; instead the secondary level $\tau = 0.90$ of the inherited grid is measured at map grade. Fully censored stairs are not solved in Stage C and are drawn as a T highland. Stage C is one pass with a mechanical completion gate; the case cap is 500,000. Stage C ran on a single machine — the same solve node that produced every map record — with the driver launching, per pair, the same fresh child process the map's transport layer launched (contract v1.2, clause S4 stage_C_single_node, adopted 2026-09-15). It halted once on an environment condition (the driver process exhausted the machine's commit limit), resumed from its own records after the page file was enlarged, re-solved nothing, and passed its 21-check gate.

Stage D is the holdout: the next 64 points of the same Sobol sequence (indices 300–363 in the zero-based prefix-stable draw whose first 300 points are the screen), decouple arm, three rungs per point at the frozen analysis plan's predicted $\hat{H}^*(x; 0.95) \pm 1$, $n = 300$ each, banked tails. A point whose pinned curve never reaches τ inside the ladder is excluded before solving (16 points; solve 0), leaving 48 solved conditions. A point predicted at the floor (predicted $h = 0$) has two rungs because $h = -1$ does not exist. The three Stage D rungs are centred on the pinned GAM's own prediction (M3). A

local reference label chosen inside that window lies within one person of its centre by construction.

This proximity concerns the observation rule and supplies no independent guarantee about a global minimum outside the window. The holdout flow is therefore 64 fixed new Sobol conditions → 16 unobserved exclusions → 48 solved conditions → a selected singleton subset. Neither the 540 promoted stairs nor that singleton subset is a random sample of real wards.

The original Stage D plan named headcount MAE, within-one-person hit rate, under-estimation rate and holdout log-loss. This revision reports headcount errors only on the selected singletons and set compatibility on the available solved conditions, without a pass line or a claim of general accuracy. Log-loss evaluates fitted probabilities against decided F/I outcomes; it is not a completion-set score and remains in the historical analysis files rather than being presented as a corrected minimum-headcount result. Stage D ran on 2026-09-19 on the same single node and passed its 23-check gate. No model was refitted after Stage D.

2.11 Analysis plan K as executed, including declared deviations

The analysis plan was frozen with Stage C/D, as a list of intended operations; code and individual prediction freezes have the distinct timing disclosed above. What was actually executed, including deviations that matter for identification, is as follows.

1. **Primary probability surface.** Logistic GAM $\text{logit } p = f(h) + \sum g_j(x_j) (+ h \times \lambda_L)$ (Hastie and Tibshirani, 1986; Wood, 2017), penalised splines on h , W , v , ρ , λ_L , j , s and factor terms on OF and P;

no monotonicity constraint on f ; additive and $h \times \lambda_L$ specifications. Fitted per arm on binomial rows with pyGAM (Servén and Brummitt, 2018). Smoothing parameters were selected over the six-point lambda grid using the default pyGAM objective for the binomial model, **UBRE**. That smoothing selection is distinct from the grouped predictive evaluation later used on the nested arm. The v0.3 wording that smoothing was selected by grouped CV is incorrect.

2. **Comparator.** Plan K named a Gaussian-process classifier (Rasmussen and Williams, 2005; scikit-learn, Pedregosa et al., 2011). Implementation deviation, declared from the analysis code: a Bernoulli-row GP over the decided cases is not computable at this scale, so the working comparator is Gaussian-process *regression on empirical logits* as a probability surrogate, with binomial variance as heteroscedastic noise. Same cells, same τ , different likelihood approximation. Alternative-model rows have different prediction availability and cannot be ranked as equal-denominator whole-design validation.

3. **Fitted-curve crossing surrogate.** The first rung at which the fitted curve reaches τ , on the ladder $h = 0-11$. If the curve never reaches τ inside the ladder the point is `boundary_not_located`.

4. **N_0/Δ decomposition**, with partial-identification language already present in the plan. That language is now enforced: Δ is not reported as a validated global share where the global minimum is not identified.

5. **Formula ladder** — $M0 = \text{round}(1.6 \times W) \rightarrow M1$ linear, five terms $\rightarrow M2$ with interactions, at most nine terms $\rightarrow M3$ the GAM surface read at τ ; forward-selected terms, with predictive evaluation kept distinct from smoothing-parameter selection. $M1$ and $M2$ were fitted on the pooled decouple stairs whose local reader had treated H^* as a number. Those training labels include invalid global-minimum tags (§3.3). Re-scoring does not repair them; no refit is undertaken in this revision. Already examined Stage D cannot become a fresh independent validation set for any future refit.

6. **Uncertainty**. Wilson intervals per cell; percentile cluster bootstrap by Sobol point ($B = 2,000$) for descriptive quantities computed from already-fixed stair labels. That bootstrap does not propagate cell-level uncertainty through a boundary reader. Simultaneous probability/surface confidence bands, and an all-slice within-one-person confidence band, were planned (K6) and **were not delivered**. Fixed-label bootstrap is therefore insufficient for population inference about H^* .

7. **Sobol decomposition** (Sobol', 1993) — first-order and total-effect indices by the Saltelli/Jansen estimators (Jansen, 1999; Saltelli et al., 2010) of a GAM-derived fitted-crossing surrogate, N_0 , and their difference. When the fitted GAM does not reach τ , the calculation substitutes $h = 12$, one beyond the highest evaluated rung $h = 11$. The resulting indices describe this surrogate-and-substitution calculation. They cannot infer the variance decomposition of an unbounded or absent true global H^* .

Items 1–3 and 7 were fitted on screen + map + Stage C decided cells. Item 5's formulas used the pooled decouple stairs whose local H^* was a number. For the decouple arm no cross-validation is

reported as holdout replacement; Stage D is the only new-condition score, and it is now a descriptive completion-set comparison. The nested arm, which has no holdout, keeps grouped five-fold CV by Sobol point and is labelled CV. The interim artefacts of v0.1 remain unchanged on disk; every final artefact carries the hash of the stair table it was computed from. The plan-K fitting code and its frozen artefacts remain unchanged. The completion-set reader is a separate post hoc analysis; publication generators and checks were updated without refitting those models.

2.12 Verification pack and corrected Anchor-002 history

This programme's second study does not rest on a replication of Paper 1. Its core computational verification pack was executed before the Stage 0a campaign. The following list distinguishes that inherited evidence from current publication checks: **(i)** specification conformity — the published rule contract is re-derived from the sealed code and the extractor fails on disagreement; **(ii)** model-fingerprint identity — the re-implemented compiler reproduces, byte for byte, the model fingerprints of all 4,050 sealed Paper-1 roots (Gate A of the anchor replication, inherited); **(iii)** differential verification — schedules that satisfy the published specification, fed as fixed assignments to the product model, are accepted by it: L0 checked 12,324 sealed feasible schedules with two independent checkers (0 disagreements; 72/72 negative-control mutations caught), L1 accepted 3,787 of 3,787 positive controls, L2 accepted 900 of 900 random and 150 of 150 targeted constructions, at 5,938 of the 6,000-record budget; **(iv)** machine independence and determinism, inherited from Paper 1's

qualification; **(v)** current publication checks, added or updated after execution and run inside the regression suite — the manuscript number checker, the forbidden-term sweep, and the table generator and full-text binding used for this revision.

Anchor history (mandatory statement). Before this design, the programme attempted to enter the successor study through a statistical replication of Paper 1's three anchor cells. ① Two attempts were made and were ended by their pre-specified stopping rules; the worst-case family-wise error rate across the two attempts, under the programme's internally pre-specified definition, is 0.00274. ② In the second attempt all three equivalence intervals lay inside the stated ± 4 pp margins. Inclusion of zero in those intervals is not a reason for failure of the equivalence criterion; the intervals satisfied the stated margins. The overall NO-GO decision remained because the additional zero-T requirement was not met, with one time-censored root in A24 and one in B32. That operational composite decision must not be restated as failure of equivalence solely because an interval included zero. ③ Re-applying Paper 1's own benchmark rule (simultaneous Clopper–Pearson bounds; Clopper and Pearson, 1934) to the new counts, B32's label moved from does not meet to indeterminate; that change came from an increase of 6 in F, independent of T. ④ No claim is made that Paper 1's results were reproduced; what was achieved is self-implementation reproduction within this programme, and Paper 1 is unchanged by any of it. A fourth difference must be stated alongside the differences that had confounded the first attempt: the model builder source differs from the sealed Paper-1 attempt (39f56daa... →

43d112dd...; the added code is gated to non-primary arms, and Gate A reproduced all 4,050 sealed model-proto hashes with the current builder), so the difference is documented and empirically bounded rather than assumed away. No third anchor attempt exists and none is planned. The full replication report is the supplement (manuscript_paper2_anchor002_v0.2.md).

2.13 Ethics

Human-subjects research: not applicable — fully synthetic scenarios; no patient, staff or identifiable data.

S4. Phase 2 — results (verbatim)

3. Results

The tables bind frozen execution reports to the separately audited post hoc completion-set analysis.

Pooled map + Stage C numbers describe the 540 promoted stairs and are in-sample for every model fitted on them. *Stage D* numbers describe the designed-to-included-to-selected-singleton flow of §2.10. Completion-set counts are not counts of population thresholds located with statistical confidence. No result below is a patient-safety finding.

3.1 Campaign execution

Table 2 records pairs, terminal states and mechanical gates for the screen waves, the map, Stage C infill and Stage D. Table 3 records screen-to-map promotion.

Table 2. Execution accounting.

segment	planned pairs	accepted pairs	F	I	T	E	gate
B screen wave 1	199764	199764	329734	22580	35373	0	PASS
B screen wave 2	135036	135036	223360	14302	24438	0	PASS
B map	713112	713112	638488	40193	12657	0	PASS
C infill	459240	459240	404273	27784	13287	0	PASS
D holdout	43292	43292	38791	2225	84	0	PASS

Pairs and arm states are distinct accounting units in the frozen reports. F/I/T denote

feasible/infeasible/time-censored outcomes; E is a fail-closed error. Conditional rates exclude T; finite-

batch censoring bands retain T, including recorded dependent cases whose banked tail did not decide.

Completion gates verify execution accounting, not scientific identification.

Table 3. Historical screen promotion.

historical selection quantity	count
stairs	600
promoted	540
boundary_not_located	46
fully_censored	14
map_cells	1382

These frozen screen labels selected subsequent windows; they do not establish global minima. Map+C

pooled promoted stairs are the primary re-reading cohort; secondary-level stairs are separate.

All segments completed with E = 0 and passed their mechanical gates. Off-day equality was re-

derived from stored feasible schedules on the map, Stage C and Stage D with zero violations, so the

OF axis moves the validator's own arithmetic, not only the compiled case. Tail-decision and current-month T fractions are those of the sealed reports; they are accounting facts, not a new raw-record audit. Campaign execution and record accounting are retained within that audited scope. The original erratum audit matched its 263 baseline files. This publication revision preserves the scientific inputs while deliberately updating manuscript generators and publication check reports; a separate preservation report lists those exceptions. The scientific correction in this revision is about what those records identify, not about whether the jobs ran.

3.2 Completion-set identification

Table 4 reports, for Stage D and for map-plus-Stage C, the numbers of stairs whose completion set is a singleton, a finite non-singleton set, or unbounded/absent, under both the conditional and the censoring band constructions.

Table 4. Finite-batch completion sets.

design subset	completion mode	total	singleton	finite multiple	unbounded / absent
D solved conditions	conditional	48	10	26	12
D solved conditions	censoring	48	10	26	12
map+C promoted stairs	conditional	540	252	72	216
map+C promoted stairs	censoring	540	240	80	220

Finite-batch logical completion sets on $h \geq 0$; no monotonicity or population confidence coverage is asserted. Missing rungs and possible nonattainment remain in the sets.

On Stage D the designed set has 64 conditions, of which 16 were unobserved exclusions and 48 were solved. Under the conditional reader the solved conditions split into 10 singletons, 26 finite non-singleton sets, and 12 unbounded or absent tails. Those are possibility-set cardinalities. They are not a count of global minima located with confidence.

On the promoted map-plus-Stage C stairs (540 stairs from a 600-stair screen), the conditional reader yields 252 singletons and the censoring reader yields 240 singletons. Unbounded or absent tails remain common under both constructions. Holes inside finite sets are preserved: a stair can have more than one possible first attaining rung because a missing lower rung, a nonmonotone dip, or a band that straddles τ prevents unique identification. An interval envelope of the candidate rungs would hide those holes and would mis-score within-one-person hits.

The v0.3 local-crossing vocabulary (located, floor-pinned, resolved, above, unresolved) remains a description of short-window behaviour. It is not reused here as a global H^* label. First observed attainment, local adjacent crossing, and the logical set of possible first-attaining finite-batch rungs are reported as distinct objects. Where the set is unbounded, nonattainment remains possible and no numerical error is assigned.

3.3 Invalid historical labels and uncorrected training targets

Table 5 records the impact of the completion-set reading on labels that v0.3 treated as exact global minima, and on the frozen M1/M2 training set.

Table 5. Historical label impact.

subset	rows	old exact labels	unsupported exact	earlier observed attainment	unsupported lower bound
D	48	21	12	3	22
map_C	540	276	61	7	188
training_decouple	144	144	33	4	33

The training row is the primary decouple-arm M1/M2 label set. These are defects in interpreting local labels as global minima even with empirical conditional rates held fixed. Re-scoring does not repair model training.

Observed counterexamples to treating a later local crossing as the first global attainment:

D condition	old reported h	earlier observed attaining h	earlier $F/(F+I)$
x318	3	1	292/300
x333	4	2	285/300
x335	6	4	289/300

A substantial share of the old Stage D point labels and of the old map-plus-Stage C labels lack support as exact global minima even if the empirical conditional rates are treated as fixed. The frozen primary M1/M2 training set has 33 affected labels among 144 decouple-arm labels. Sample selection

is explicit: formulas were fitted on stairs the local reader had already converted into a number, a selected and floor-heavy subset of the design. Those invalid training labels remain in the frozen artefacts. Re-scoring alone does not repair training-target validity, and this revision does not refit.

3.4 Descriptive re-score of frozen predictors

Table 6 reports singleton MAE and within-one-person hits on the selected Stage D singletons. Table 7 reports all/some/none set hits of each frozen predictor against each condition's candidate set, without assigning a numerical error to possible nonattainment. Alternative-model rows (including the two GAM specifications and the empirical-logit GP) have different prediction availability; excluded points and unavailable predictions are not silently imputed as successes or failures.

Table 6. Frozen predictions on selected D singletons.

predictor	available D / 48	singleton n	within ± 1	MAE (people)
M0	48	10	2	2.90
M1	48	10	9	0.70
M2	48	10	9	0.50
M3	48	10	10	0.40
N0_posthoc	48	10	9	0.30
K:gam_h_x_lambda_L	47	10	10	0.40
K:gp_logit	37	10	10	0.10

N0_posthoc is the arithmetic-floor comparator introduced during this correction. M3 duplicates

K:gam_additive. The selected singleton subset is floor-heavy; unavailable predictions and excluded

design points are not scored. These results establish no whole-design or population performance ranking.

Table 7. Censoring-set compatibility of frozen D predictions.

predictor	available	all completions	some completions	no completions
M0	48	4	19	25
M1	48	10	34	4
M2	48	11	33	4
M3	48	10	27	11
N0_posthoc	48	13	33	2
K:gam_h_x_lambda_L	47	10	27	10
K:gp_logit	37	13	15	9

Each column classifies ± 1 compatibility with the finite-batch candidate set, retaining its holes and possible nonattainment. These are not confidence bounds on prediction accuracy.

On the selected Stage D singletons, M1 records 9 within-one-person hits with MAE 0.7; M2 records 9 hits with MAE 0.5; the post hoc N_0 comparator records 9 hits with MAE 0.3; and M0 records 2 hits with MAE 2.9. The selected singleton subset includes 8 floor-associated singletons. M1/M2 and the arithmetic N_0 comparator have the same singleton within-one-person hit count in this selected subset; N_0 also has the lower singleton MAE here. That pattern does not establish global superiority of any predictor. The old 67% general-accuracy claim cannot be replaced by a new percentage on this subset and offered as a whole-design result.

M3 selected the observation window. Apparent singleton proximity of M3 is therefore not independent evidence that its surface estimates global minima accurately. All/some/none columns describe each point relative to its candidate set; they are not confidence bounds on population accuracy. Direction below all finite candidates is reported only where a finite minimum is assumed; nonattainment still carries no numerical error.

3.5 Arithmetic comparator against N_0

Table 8 reports the arithmetic comparison of $M0 = \text{round}(1.6 W)$ and of post hoc $\text{ceil}(1.6 W)$ against N_0 on the promoted design.

Table 8. Arithmetic floor comparisons.

arithmetic comparison	stairs	denominator	percent
$\text{round}(1.6W) < N_0$	351	540	65.0
$\text{ceil}(1.6W) < N_0$ (post hoc)	307	540	56.9

These compare headcounts with the compiler's arithmetic floor over promoted synthetic stairs. They do not measure patient safety, annual FTE, or a validated global minimum.

On the 540 promoted stairs, $\text{round}(1.6 W)$ lies below N_0 on 351 stairs and $\text{ceil}(1.6 W)$ lies below N_0 on 307 stairs. These are arithmetic statements over the promoted design, not patient-safety findings and not boundary-dependent claims. They separate a counting deficit of the 1.6 rule of thumb from any claim about located H^* . The superseded v0.3 statement that the multiplier missed a located floor on

almost the entire design mixed those two things. The present tables keep them apart. Where M0 lies below every finite candidate of a completion set, that is a descriptive set-relation, not a definite population underestimation of a global minimum.

3.6 Deterministic band sensitivity

Table 9 reports how singleton counts and finite upper endpoints change when every cell band is widened by a deterministic $\pm$ -epsilon grid while τ stays fixed.

Table 9. Deterministic band-widening sensitivity.

subset	epsilon	singleton count	count with finite upper endpoint
D	0.0	10	36
D	0.005	7	27
D	0.01	6	22
D	0.02	3	12
map_C	0.0	240	320
map_C	0.005	148	248
map_C	0.01	94	180
map_C	0.02	29	62

Post hoc deterministic widening of the censoring bands by $\pm$ epsilon, clipped to $[0,1]$, with tau fixed at 19/20. Epsilon is neither an estimated standard error nor a coverage-calibrated confidence bound.

The perturbation grid is post hoc and is not selected to achieve a favourable result. No epsilon is a standard error, a coverage level, or an externally justified uncertainty bound. The grid is not a change of τ and is not estimated sampling error. It shows only that unique attainment is sensitive to how

wide one is willing to draw a logical band around a finite-batch rate. That sensitivity is a reason not to treat singleton counts as population thresholds.

3.7 Plan deviations that remain

Table 10 records the inferential deliveries that plan K named and that this campaign did not provide, together with the sampling features that additional inference would have to confront.

Table 10. Deviations and interpretive limits.

item	record / deviation	current interpretation
Global boundary reading	Local adjacent upward crossings used as global minima	Separate observed attainment, crossings and logical completion sets
Training labels	Primary M1/M2 used local labels	Retain frozen coefficients; disclose affected labels; no superiority claim
D chronology	GAM table/run freeze before execution; M1/M2 ladder and reader during execution	No external registration or complete blinding claim; monitor exposed x347 h4, x348 h0/h1
K6 uncertainty	Fixed-label bootstrap and cell Wilson summaries	No simultaneous population surface or slice coverage established
Smoothing selection	Six-lambda pyGAM gridsearch	Default binomial UBRE; grouped predictive evaluation is separate
Sobol target	Fitted GAM crossing; absent crossings substituted by h=12	Surrogate-and-substitution sensitivity only; no global H* decomposition
Anchor-002	Equivalence margin satisfied; extra zero-T gate failed	Retain composite NO-GO; zero inclusion is not equivalence failure
New comparators / sensitivity	N0, ceiling and epsilon grid introduced after D	Explicitly post hoc, descriptive and not refitted or confirmatory

This analysis deliberately separates (a) logical completion of finite-batch censored outcomes, (b) deterministic probability-band perturbation, and (c) population sampling inference, which is not delivered here. Simultaneous surface bands were not delivered. The cluster bootstrap of already-fixed

labels does not close that gap. Additional inferential work would require a defined target, tail-bank/root assignment units, handling of map/C outcome-adaptive sampling, and propagation through minima and model fitting. The present aggregate-only correction neither supplies nor claims those guarantees. Outcome-adaptive Stage C selection, banked-tail dependence, unobserved rungs, and the M3-centred Stage D window all remain.

3.8 Fitted surfaces as originally estimated

Table 11 describes the frozen prediction mechanisms and their interpretation limits. Historical log-loss and local-label performance remain in the archived analysis files rather than being presented as corrected global-minimum scores.

Table 11. Frozen prediction mechanisms.

label	prediction mechanism	scope
M0	round(1.6W)	Fixed arithmetic comparator; original Python rounding
M1	Frozen regression for staffing ratio	Primary decouple local-label training; no lower-bound guarantee
M2	Frozen regression for premium above N0	Same affected training-label issue
M3	Additive logistic GAM probability surface	Selected D observation windows; UBRE smoothing selection
Interaction GAM	GAM with h by lambda_L interaction	Alternative frozen probability surface
GP logit	Gaussian-process regression on empirical logits	Probability surrogate; incomplete D prediction availability
N0_posthoc	Compiler arithmetic floor	Added descriptive comparator; not a fitted model

Those surfaces remain what was fitted. They are not repaired by the completion reader. Partial dependence on h that saturates after a few rungs above N_0 is qualitative context for why $\tau = 0.95$ is hard to identify: adding heads moves a decided rate that already sits near the target by very little, and nonmonotone dips are compatible with that plateau. Disagreement between GAM and GP about the first rung at which a fitted curve reaches τ is expected on a plateau and is not resolved by declaring one family the winner. Sobol ranks derived from the GAM-plus- $h=12$ substitute are reported, if at all, only as ranks of that surrogate calculation (source manifest). They are not a variance decomposition of true global H^* .

3.9 Reader coverage under designed truth

Table 12 reports the controlled validation of §2.6 (design in that section): 195 configurations of designed curve $\times$ true first attainment $\times$ window offset $\times$ censoring share, 2000 replications each, $n = 300$ per rung, with both readers imported unchanged.

Table 12. Completion-set reader and v0.3 local rule under designed truth.

designed curve	configurations	coverage, conditional (%)	coverage, censoring (%)	coverage, censoring $\pm 1/100$ (%)	singleton (%)	singleton and correct (%)	old rule located (%)	old rule located and correct (%)
dip after attainment (0.97, 0.93, 0.97)	45	97.9	98.9	99.9	26.2	25.9	61.9	25.8
never attains (0.94)	15	42.0	76.0	91.8	0.0	0.0	37.6	0.0

plateau near τ (0.945 $\rightarrow$ 0.952)	45	54.6	71.2	89.2	27.8	15.6	46.3	9.7
ramp (+0.006 per rung through τ)	45	56.1	71.0	89.2	28.1	14.9	48.6	9.3
step (0.90 $\rightarrow$ 0.97)	45	98.0	99.0	99.9	26.7	25.9	29.1	25.9

By window offset ($h^* = 3$ and 6, where every offset is realisable without floor clipping):

window centre – true h^*	configurations	coverage, conditional (%)	coverage, censoring (%)	singleton (%)	singleton and correct (%)	old rule located and correct (%)
+0	24	66.3	78.1	0.0	0.0	66.3
+1	24	77.2	83.0	0.0	0.0	0.0
+2	24	100.0	100.0	0.0	0.0	0.0

Designed generator curves on $h = 0\text{--}12$, true first attainment $h^* \in \{0, 3, 6\}$, three-rung windows

centred at $h^* + \text{offset}$, $n = 300$ per rung, censoring shares 0, 0.02 and 0.10, 2000 replications per

configuration (seed 20260920); means over configurations with equal weight. Coverage = the true first

attainment lies in the completion set (for the never-attaining curve: nonattainment kept possible).

Readers are imported unchanged from the sealed erratum and from the v0.3 map reader. Designed

curves are not ward populations and the frozen generator was not run.

When the designed rates sit well away from τ (step and dip-after-attainment curves) the conditional

completion set contains the true first attainment in 98.0% and 97.9% of batches; the misses are

attaining rungs whose observed rate fell below τ by sampling. When every rung lies within one standard error of τ (plateau 54.6%, ramp 56.1%) or the curve never attains (42.0%), conditional coverage collapses, because a rate observed on the wrong side of τ is read as certain. The censoring band recovers part of it only when censoring is present (71.2% and 76.0% overall), and the $\pm 1/100$ widening reaches 89.2% and 91.8%, at the cost of larger sets. Singletons arise only when the window reaches the floor, so that no lower rung is left unobserved — here the $h^* = 0$ configurations and the windows centred two rungs below $h^* = 3$; averaged over all configurations that is roughly one batch in four for every attaining curve. On the step curve a singleton names the true rung 97.1% of the time, on the plateau curve 56.1%. The offset rows ($h^* = 3$ and 6, where every offset is realisable without floor clipping) show the mechanism: a window centred two rungs below the truth yields a singleton in 16.8% of batches and that singleton is correct in 0.0% of them, because it can only arise from a below- τ rung observed above τ ; a window centred two rungs above the truth is always covered (100.0%) and never a singleton (0.0%), because the lower rungs are unobserved. Under 10% censoring the censoring band's coverage rises to 92.4% while its singleton rate falls to 0.0%. The v0.3 local rule locates a rung in 29.1% of step batches and, when it does, names the true one 88.8% of the time; on the plateau curve the located rung is correct 20.9% of the time.

These numbers quantify, for designed curves only, the gap that §2.6 states in words: the completion set is complete for the observed rates and is not a confidence set for the generator rates. They do

not transfer to the frozen generator, to the already examined Stage D conditions, or to any ward population, and they do not license a coverage statement for Table 4.

S5. Phase 2 — discussion (verbatim)

4. Discussion

4.1 Scientific contribution

The contribution of this paper is a reproducible computational experiment, under a published hard-rule specification, that records finite-batch feasibility along headcount stairs and then states, without over-claim, what those records can identify. Arithmetic N_0 remains a solver-free counting floor. Finite-batch F/I/T counts remain machine-checked observations. The completion-set reader converts those observations into conservative possibility sets that keep holes, unbounded tails and absent minima. That is a methodological result about identification in synthetic rostering measurement. It is not a validated staffing formula, and it is not a demonstration that a five-term regression outperforms a multiplier on a population of wards.

Narrowing the claim is itself the correction. A local crossing on a three-rung window, a first observed attainment in a finite batch, and a global minimum of a generator probability are three objects. The campaign records the first two; the descriptive correction does not establish the third. Incompleteness of finite-batch first-attainment sets and uncertainty about a generator probability are separate limitations. Making that distinction explicit is the scientific service of v0.4.

4.2 Synthetic-design assumptions

Results hold for the frozen generator, the published rule contract, the inherited time ladder, banked prior-month tails, a 30-day month, one manager, full-time staff, and the eight-axis box. The nested and decouple arms are stylised composition paths, not ward populations. W is an externally specified daily demand total, not an estimated workload. Headcount is an integer roster size, not FTE and not a patient-safety threshold. Domain $h \geq 0$ inherits the compiler's N_0 premise; unprobed admissibility is not proved. Within-cell tail sharing introduces a shared input; Wilson intervals do not propagate tail-bank uncertainty. Map and Stage C sampling is outcome-adaptive. Stage D conditions are the next prefix of a fixed Sobol sequence, not a random draw from real wards. The generator probability is not a clinical event probability. Real-ward studies fit one hospital's nurse profiles, requests and forecast demand over a few months — for example the 60-nurse, five-month surgical-ward case of Zhang, Tang and Lau (2025) — whereas this campaign sweeps a synthetic eight-axis box with a frozen generator; neither design stands in for the other. Supply-side uncertainty such as absence is outside the frozen generator, as it is a stated extension in that demand-uncertainty literature (Zhang, Tang and Lau, 2025). None of these assumptions is hidden, and none of them is removed by re-reading the counts.

4.3 Verification strengths

Within audited scope the campaign is strong as computation. The rule specification is independent of the product code and is re-derived by an extractor that fails on disagreement. Gate A reproduced all

4,050 sealed Paper-1 model fingerprints. Differential verification accepted the positive controls and caught the negative-control mutations at the recorded counts (§2.12). Terminal E is fail-closed and was never observed. Censoring is recorded and is not imputed. N_0 is computed without a solver and is inherited as a working-domain premise; the absence of recorded F below it is a consistency check. Stage gates are mechanical. Tables in this file are bound to artefacts by generated blocks rather than by hand transcription of current results. Those strengths support a descriptive computational paper. They do not promote finite-batch labels into population minima.

4.4 Prior failed anchors

The programme did not enter this design by reproducing Paper 1. The two Anchor-002 attempts stopped under their pre-specified rules, with internally defined family-wise error 0.00274. Equivalence intervals in the second attempt satisfied the stated ± 4 pp margins; zero inclusion is not failure of that criterion. Overall NO-GO remained because of the extra $T = 0$ gate. Re-application of Paper 1's Clopper–Pearson benchmark moved one cell label independently of T. No claim is made that Paper 1 was reproduced. Self-implementation reproduction inside this programme, plus the documented builder-hash difference bounded by Gate A, is what was achieved. That history is retained because failure to satisfy the full anchor-replication gate cannot be converted into external validation of a later staffing surface.

4.5 Identification limits and superseded claims

Nonmonotone observed rates, missing lower rungs and prediction-centred windows often leave more than one possible first attainment of the reported finite-batch rates. The completion sets display this descriptive ambiguity as holes, finite non-singleton sets and unbounded tails with possible nonattainment. A separate inferential argument would be required to draw conclusions about identification of the population $H^*(x; 0.95)$. Conditional q and the all-generated probability remain distinct; synthesized T stays inside the censoring band. Simultaneous confidence bands of the surface were not delivered, so this is not a population confidence analysis. The designed-truth check of §3.9 shows the size of that gap: coverage of the true first attainment is near-complete when rates sit well away from τ and falls to 54.6% (conditional) when every rung lies within one standard error of τ , the regime in which decided rates that already sit near the target move very little per added head (§3.8).

Several v0.3 headlines are therefore superseded and are not primary current results. The claim that about 91% of the requirement is arithmetic and 9% combinatorial depended on invalid global labels and on surrogate outcomes; it is not established for global H^* . The claim that the 1.6 multiplier missed a located floor on 96% of stairs mixed an arithmetic N_0 deficit with a boundary-dependent statement; the current primary comparison is the arithmetic count 351 (round) and 307 (ceil) versus N_0 , not a recycled 96% underestimation of H^* . The claim of 67% general within-one-person accuracy for a five-term formula on holdout points does not survive: the reference sets have changed, the

training labels are partly unsupported, the scored subset is a selected singleton collection, and M3 centred the window. Those historical percentages may be found in sealed v0.3; they are not restated here as findings.

M1 is an unconstrained regression prediction. It is not a lower bound and not a staffing guarantee.

M3's apparent hit rate is window-centred and is not proof of best predictive performance. Sobol ranks describe a GAM surrogate with $h = 12$ substitution after $h = 11$; they do not identify the variance decomposition of an unbounded or absent true minimum.

4.6 Practical limitations and non-claims

This study does not estimate safe, adequate, optimal or minimum staffing for any real ward. It does not estimate a legal standard, an FTE budget, or a patient-outcome threshold. It does not claim that any real ward's roster is infeasible at $1.6 \times W$. It does not claim that the frozen ladder is ready for use. It makes no algorithmic performance claim and its results are not comparable to rostering-benchmark scores. It does not claim external registration, complete blinding, or a new raw-record audit. It does not claim that re-scoring repaired M1/M2 training targets. It does not claim that already examined Stage D can serve as a fresh validation set after a future refit. Alternative predictors cannot be ranked as if they had equal availability. Fixed Sobol points do not generalise to a population of wards.

The v0.3 "conditions of use" box that would have printed the five-term formula as a lower bound for a roster to exist is withdrawn. Frozen coefficients remain historical artefacts in the sealed ladder file. Printing them as a tool would over-claim the identification that this revision exists to retract.

4.7 What the numbers look like on the multiplier scale

Readers who arrive with the practical question — not 1.6, then what? — deserve a descriptive answer on the scale they use, with the same limits attached. Over the 540 promoted stairs ($W = 12\text{--}27$ people on duty per day), the solver-free arithmetic floor alone is $N_0 / W = 1.67 \times W$ at the median (interquartile 1.58–1.77, 90th percentile 1.84, range 1.46–1.93); that spread is produced deterministically by the night cap, the off-day entitlement, leave load and role coverage of each condition, which is why $\text{round}(1.6 W)$ falls below the counting floor on 351 of 540 stairs (§3.5). On the 324 stairs where some observed rung reached τ conditionally, the first such rung was $1.79 \times W$ at the median (interquartile 1.71–1.86, 90th percentile 1.92, range 1.54–2.18); on the remaining 216 stairs no rung inside the observed window reached τ , so the headcount that would be larger than the window and is not reported. Where attainment was observed it was at the floor itself on 173 of the 324 stairs and within three heads above it on 278, the largest observed premium being 9 heads. These are first *observed* attainments of finite batches, not identified minima (§2.6, §3.2), inside one synthetic generator and one rule contract, and they describe the box, not any ward. What they do show is the shape of the answer: a single universal multiplier is the wrong object, because most of the variation is

already fixed by counting a ward's own rules, with a combinatorial premium that was zero to three heads above that floor on most stairs where attainment was observed — and unobserved, hence larger than the window, on the rest. Turning that shape into a usable table — the multiplier as a function of rule parameters, measured on full stairs with cell sizes that resolve rates near τ , and checked against real wards — is the successor study, not a corollary of this one.

4.8 Reproducibility

The execution contract, freeze and report artefacts, descriptive and final-analysis artefacts (interim ones retained), the completion-set analysis of 2026-09-20, and the checkers that bind generated tables are in the study package. Per-evaluation records and the pinned rostering software source are retained institutionally and are not redistributed. The published rule specification allows a third party to pose the same feasibility problem to their own solver. Public repository URL: [TBD]. The dated SCIENTIFIC_ADDENDUM.txt is the historical measurement basis; its proposed title and statements that prose had not yet been inserted describe the earlier revision stage, not this manuscript. Current completion status and review disposition are recorded in revision_v04/final_review/. Reproduction of the identification correction is the erratum pipeline already run on the sealed hashes recorded in summary.json; scientific inputs and the erratum analysis remain unchanged. Publication generators, checkers and their tests were updated, with the permitted exceptions recorded in the preservation

report. Verification remains self-implementation reproduction within this programme; no external party has audited the pipeline.